

Naveen Jindal School of Management

Computer Information Systems and Technology (BS)

Bachelor of Science in Computer Information Systems and Technology

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

Faculty

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Professors: Ashiq Ali, Alain Bensoussan, Gary Bolton, Metin Çakanyildirim, Huseyin Cavusoglu, Jianqing Chen, William M. Cready, Gregory G. Dess, Umit G. Gurun, Dorothée Honhon, Kyle Hyndman, Varghese S. Jacob, Sanjay Jain, Ganesh Janakiraman, Elena Katok, Dmitri Kuksov, Nanda Kumar, Seung-Hyun Lee, Stanley Liebowitz, Zhiang (John) Lin, Sumit K. Majumdar, Stanimir Markov, Amit Mehra, Syam Menon, Vijay S. Mookerjee, B. P. S. Murthi, Vikram Nanda, Özalp Özer, Mike W. Peng, Hasan Pirkul, Cuili Qian, Suresh Radhakrishnan, Srinivasan Raghunathan, Ram C. Rao, Brian Ratchford, Michael J. Rebello, Gil Sadka, Sumit Sarkar, Suresh P. Sethi, Kathryn E. Steckle, Riki Takeuchi, Wing Kwong (Eric) Tsang, Jun Xia, Ying Xie, Harold Zhang, Zhiqiang (Eric) Zheng

Associate Professors: Mehmet Ayvaci, Nina Baranchuk, Zhonglan Dai, Rebecca Files, Michael Hasler, Bin Hu, Surya N. Janakiraman, Robert L. Kieschnick Jr., Atanu Lahiri, Jun Li, Ningzhong Li, Maria Loumiotis, Livia Markóczy, Ramachandran (Ram) Natarajan, Naim Bugra Ozel, H. Dennis Park, Anyan Qi, Young U. Ryu, Serdar Simsek, Harpreet Singh, Upender Subramanian, Shaojie Tang, Shouqiang Wang, Kelsey D. Wei, Han (Victor) Xia, Yexiao Xu, Alejandro Zentner, Jieying Zhang, Yuan Zhang, Zhe (James) Zhang, Feng Zhao, Yibin Zhou

Assistant Professors: Khai Chiong, Rafael Copat, Soraya Fatehi, Andrew Frazelle, Ying Huang, Joonhwi Joo, Sora Jun, Jason Kautz, Tongil Kim, Sheen Levine, Christopher Mace, Samir Mamadehussene, Jean-Marie Meier, Zixuan Meng, Radha Mookerjee, Jedson Pinto, Ignacio Rios Uribe, Alejandro Rivera Mesias, Simon Siegenthaler, Kirti Sinha, Shujing Sun, Xiaoxiao Tang, Shervin Tehrani, Ashwin Venkataraman, Christian Von Drathen, Guihua Wang, Hongchang Wang, Pingle Wang, Junfeng Wu, Steven Xiao

Professor Emeritus: R. Chandrasekaran

Associate Professor Emeritus: David J. Springate

Clinical Professors: John Barden, Britt Berrett, Abhijit Biswas, Shawn Carraher, Larry Chasteen, Paul Convery, Tevfik Dalgic, Howard Dover, John Gamino, Randall S. Guttery, William Hefley, Robert Hicks, Robert Kaiser, Marilyn Kaplan, Van Latham, Sonia Leach, Peter Lewin, Jeffrey Manzi, John McCracken, Diane McNulty, Larry Norton, Divakar Rajamani, Daniel Rajaratnam, Kannan Ramanathan, Prakash Shrivastava, Mark Thouin, McClain Watson, Jeff Weekley, Habte Woldu, Fang Wu

Clinical Associate Professors: Shawn Alborz, Dawn Owens, Carolyn Reichert, Avanti P. Sethi, Ramesh Subramoniam, Aysegul Toptal, David Widdifield

Clinical Assistant Professors: Athena Alimirzaei, Christina (Krysta) Betanzos, Moran Blueshtein, Jeffery (Jeff) Hicks, Dupinderjeet Kaur, Revansiddha Khanapure, Kristen Lawson, Liping Ma, Parneet Pahwa, Jason Parker

Professors of Instruction: Semiramis Amirpour, Mary Beth Goodrich, Chris Linsteadt, Suzette Plaisance Bryan, Matt Polze, Luell (Lou) Thompson

Associate Professors of Instruction: Judd Bradbury, Monica E. Brussolo, Ayfer Gurun, Maria Hasenhuttl, Julie Haworth, Thomas (Tom) Henderson, Kathryn Lookadoo, Sarah Moore, Hirofumi Nishi, Daniel Sibley, Agnieszka Skuza, Hubert Zydorek, Mohammad aseri Taheri

Assistant Professors of Instruction: Negin Enayaty Ahangar, Daniel Karnuta, Joseph Mauriello, Victoria McCrady, Rasoul Ramezani, Gaurav Shekhar

Professors of Practice: Gregory Ballew, Tiffany A. Bortz, Ranavir Bose, Alexander Edsel, Rajiv Shah, Donald Taylor, Keith Thurgood

Associate Professors of Practice: Nozar Hassanzadeh, Jackie Kimzey, Julie Lynch, David Parks, Margaret Smallwood, Steven Solcher, Kathy Zolton

Assistant Professors of Practice: Khatereh Ahadi, Steven Haynes, Abu Naser Islam, Edward Meda, Paul Nichols, Timothy Stephens, Guido Tirone, Robert Wright

I. Core Curriculum Requirements: 42 semester credit hours¹

Communication: 6 semester credit hours

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

Mathematics: 3 semester credit hours

[MATH 1325](#) Applied Calculus I^{2, 3, 4, 5}

Or select any 3 semester credit hours from [Mathematics Core](#) courses⁴ (see advisor)

Life and Physical Sciences: 6 semester credit hours

Select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor and degree requirements)

Language, Philosophy and Culture: 3 semester credit hours

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

Creative Arts: 3 semester credit hours

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

American History: 6 semester credit hours

Select any 6 semester credit hours from [American History Core](#) courses (see advisor)

Government/Political Science: 6 semester credit hours

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

Social and Behavioral Sciences: 3 semester credit hours

Choose one of the following:⁶_—

[BA 1310](#) Making Choices in Free Market Systems^{2, 3}_{— —}

[BA 1320](#) Business in a Global World^{2, 3}_{— —}

[ECON 2301](#) Principles of Macroeconomics^{2, 3}_{— —}

[ECON 2302](#) Principles of Microeconomics^{2, 3}_{— —}

Or select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

Component Area Option: 6 semester credit hours

Choose two of the following:⁶_—

[MATH 1326](#) Applied Calculus II^{2, 3, 4}_{— — —}

[BA 1310](#) Making Choices in Free Market Systems^{2, 3}_{— —}

[BA 1320](#) Business in a Global World^{2, 3}_{— —}

[ECON 2301](#) Principles of Macroeconomics^{2, 3} _{_}

[ECON 2302](#) Principles of Microeconomics^{2, 3} _{_}

Or select any 6 semester credit hours from [Component Area Option Core](#) courses (see advisor)

II. Major Requirements: 75-78 semester credit hours

Students must meet all course prerequisites.

Major Preparatory Courses: 21-24 semester credit hours beyond Core Curriculum

[ACCT 2301](#) Introductory Financial Accounting² _{_}

[ACCT 2302](#) Introductory Management Accounting² _{_}

[BLAW 2301](#) Business and Public Law² _{_}

[ITSS 3311](#) Introduction to Programming⁷ _{_}

[ITSS 4381](#) Object Oriented Programming with Python^{8, 9} _{_}

or [ITSS 3312](#) Object-Oriented Programming^{8, 9} _{_}

[MATH 1325](#) Applied Calculus I^{2, 3, 4, 5} _{_ _}

[MATH 1326](#) Applied Calculus II^{2, 3, 4, 10} _{_ _}

or [OPRE 3340](#) Advanced Business Quantitative Methods^{2, 11} _{_}

[OPRE 3333](#) Quantitative Business Analysis^{2, 10} _{_}

or [MATH 2333](#) Matrices, Vectors, and Data^{2, 12} _{_}

[OPRE 3360](#) Managerial Methods in Decision Making Under Uncertainty

or [STAT 3360](#) Probability and Statistics for Management and Economics

Choose two of the following:⁶ _{_}

[BA 1310](#) Making Choices in Free Market Systems^{2, 3} _{_}

[BA 1320](#) Business in a Global World^{2, 3} _{_}

[ECON 2301](#) Principles of Macroeconomics^{[2](#), [3](#)}

[ECON 2302](#) Principles of Microeconomics^{[2](#), [3](#)}

Major Core Courses: 24 semester credit hours

[BCOM 1300](#) Professional Communication^{[13](#)}

or [BCOM 3300](#) Business Communication^{[13](#)}

[BCOM 4300](#) Advanced Professional Communication

[FIN 3320](#) Business Finance

[IMS 3310](#) International Business

[ITSS 3300](#) Information Technology for Business

[MKT 3300](#) Principles of Marketing

[OBHR 3310](#) Organizational Behavior

or [OBHR 3330](#) Introduction to Human Resource Management

[OPRE 3310](#) Supply Chain and Operations Management

Students are required to complete 100 hours of community engagement:

[BA 4095](#) Social Sector Engagement and Community Outreach Practicum^{[14](#)}

Major Related Courses: 21 semester credit hours

[ITSS 4300](#) Database Fundamentals

[ITSS 4330](#) Systems Analysis and Design

[ITSS 4351](#) Foundations of Business Intelligence

[ITSS 4360](#) Network and Information Security

[ITSS 4370](#) Information Technology Infrastructure

[ITSS 4382](#) Applied Artificial Intelligence/Machine Learning

[ITSS 4395](#) Capstone Senior Project - Information Systems

or [BPS 4395](#) Capstone Senior Project - Business

A practicum experience of at least 160 working hours is required, with registration in one of the following courses.

[ITSS 4090](#) Information Technology and Systems Internship¹⁵

or [BA 4090](#) Management Internship^{15, 16}

Guided Electives: 9 semester credit hours

The BS CIST degree program requires students to focus on a specific concentration to obtain in-depth knowledge in a particular business area.

A change to a student's concentration will require a formal change in major.

Students may choose 9 semester credit hours from one of the following Concentrations.

A. Concentration in Business Intelligence and Analytics

Students who wish to focus on analytical applications on a firm technical foundation can choose the Concentration in Business Intelligence and Analytics, which requires a choice of three of the following courses for a total of 9 semester credit hours:

[ITSS 4352](#) Introduction to Web Analytics

[ITSS 4353](#) Business Analytics

[ITSS 4354](#) Advanced Big Data Analytics

[ITSS 4355](#) Data Visualization

B. Concentration in Enterprise Systems

Students who wish to focus on the internal, integrated information systems within an organization must take the following courses for a total 9 semester credit hours:

[ITSS 4340](#) Enterprise Resource Planning

[ITSS 4343](#) Integrated SCM Information Systems

or [ITSS 4344](#) CRM using Salesforce

[ACCT 4342](#) Accounting Information Systems and Financial Reporting

or [ACCT 3322](#) Integrated Accounting Information Systems

C. Concentration in IT Sales Engineering

Students who wish to focus on sales of technical products and services, must take the following

courses for a total 9 semester credit hours:

[ITSS 4340](#) Enterprise Resource Planning

[MKT 3330](#) Introduction to Professional Selling

[MKT 4331](#) Digital Prospecting

or [MKT 4332](#) Advanced Professional Sales

D. Concentration in IT Innovation and Entrepreneurship

Students who wish to focus development of new technical products in a start-up environment, must take the following courses for a total 9 semester credit hours:

[ENTP 3301](#) Innovation and Entrepreneurship

[ENTP 4330](#) Entrepreneurial Marketing

or [ENTP 4311](#) Entrepreneurial Strategy and Business Models

[ITSS 4344](#) CRM using Salesforce

or [BLAW 4305](#) Introduction to Patent Law

E. Concentration in Cybersecurity Management

Students who wish to focus on cybersecurity within a firm technical foundation can choose the Concentration in Cybersecurity, which requires the three following courses for a total of 9 semester credit hours:

[ITSS 4356](#) Data Governance

[ITSS 4361](#) Information Technology Cybersecurity

[ITSS 4362](#) Cybersecurity Governance

F. Concentration in Information Technology and Systems

Any 9 semester credit hours of ITSS upper-division courses, excluding [ITSS 4301](#) Database Systems ¹⁷ and [ITSS 4V90](#) ITSS Internship, that are not part of the BS CIST major preparatory, major core, or major related courses may be used to satisfy the 9 semester credit hours of guided elective credit.

III. Elective Requirements: 0-3 semester credit hours

Free Electives: 0-3 semester credit hours¹¹

Both lower- and upper-division courses may count as electives.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.
2. Indicates a prerequisite class to be completed before enrolling for upper-division classes.
3. A required Major course that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.
4. MATH 1325 is required for all JSOM undergraduate students. Students may elect to substitute MATH 2413 or MATH 2417 for MATH 1325
5. To make timely degree progress, students should complete MATH 1325 or its equivalent by the end of their first semester at UT Dallas. Students who will not meet this requirement should contact their academic advisor to discuss their degree timeline.
6. Certain courses listed are prerequisites for major core, major concentration, or major related courses. Choose accordingly.
7. Students who have completed CS 1436 with a C or better may replace ITSS 3311 with an additional ITSS upper-division course (excluding ITSS 4301 and ITSS 4V90).
8. Students who have completed CS 1337 with a C or better may replace ITSS 3312 with an additional ITSS upper-division course (excluding ITSS 4301 and ITSS 4V90).
9. ITSS 4381 can satisfy the ITSS 3312 major preparatory requirement in prior BS CIST catalogs if the course is not used elsewhere on the degree plan.
10. Students who choose MATH 1326 or MATH 2414 or MATH 2419 instead of OPRE 3340 will require a total of 3 semester credit hours of free electives to meet the 120 semester credit hour requirement for graduation. Students who choose OPRE 3340 instead of MATH 1326 or MATH 2414 or MATH 2419 will require another 090 Component Area Option Core Curriculum course and a total of 3 semester credit hours of free electives to meet the 120 semester credit hour requirement for graduation.
11. BPS 4395 can be taken instead of ITSS 4395 only if ITSS 4395 Capstone is not available.
12. Students may substitute MATH 2418 or CS 2305 for OPRE 3333 or MATH 2333.
13. JSOM first-time-in-college freshmen are required to take BCOM 1300 in their first semester. Transfer students and students new to JSOM are required to take BCOM 3300 in their first semester.
14. Students may fulfill the community engagement requirement with BA 4095, BPS 4396, ENTP

4340, or MKT 4360. The zero semester credit hour course BA 4095 is recommended as the most efficient way to satisfy this requirement.

15. Students may fulfill the internship requirement with ITSS 4090, BA 4090, or ITSS 4V90 (1-3 semester credit hours). The zero semester credit hour courses ITSS 4090 or BA 4090 are recommended as the most efficient way to satisfy this requirement.

16. Students employed in full-time jobs may fulfill their internship requirement with their current employer as a special project with preapproval from their program director.

17. ACCT 4301 or ITSS 4301 may not be used to satisfy BS CIST degree requirements.

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